Passenger-Response-Aware Bidirectional Meeting-Point DRT Service Design for Many-to-Many Operations

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Abstract

Many-to-many demand-responsive transit (DRT) systems must balance route consolidation, passenger convenience, and fleet responsiveness under uncertain request arrivals. Meeting-point service designs can reduce routing intensity by consolidating passenger pickups and dropoffs, but they also expose operators to acceptance losses when walking, waiting, or in-vehicle burdens make an offer unattractive. This paper studies bidirectional meeting-point DRT service design in which both pickup and dropoff locations are selected from candidate meeting-point sets, passenger response is represented through single-offer binary-logit acceptance, and vehicle routes are updated through rolling-horizon operations.

We develop a passenger-response-aware simulation framework that links service offer generation, binary-logit passenger response, and rolling-horizon dispatch. The framework is used to compare door-to-door service, single-sided meeting-point service, bidirectional service without choice, and the full passenger-response-aware design under tested synthetic service-design condi-

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tions. The formal evidence is interpreted as conditional operational evidence: bidirectional pickup and dropoff consolidation can lower vehicle-kilometres per served trip when accepted requests are spatially compatible, while also reducing served share and shifting rejection between passenger-choice and routing-feasibility channels. Diagnostic coverage controls, fixed-accepted-set routing decompositions, robustness checks, passenger-type heterogeneity analyses, post-hoc welfare accounting, and algorithm diagnostics are treated separately from the primary behavioral comparison.

The results support a logistics and operations reading of meeting-point DRT as a service-design trade-off rather than an unconditional improvement. Under the tested conditions, passenger-response-aware bidirectional meeting-point consolidation can reduce routing intensity per served trip, but operators must monitor coverage loss, choice rejection, feasibility rejection, and simulated passenger-type heterogeneity. The study therefore contributes operational service-design evidence for many-to-many DRT systems and clarifies the evidence boundaries for future empirical calibration and public-data validation.
Keywords: demand-responsive transit, meeting points, passenger choice, rolling horizon, service design, logistics

1. Introduction

Demand-responsive transit (DRT) provides flexible passenger service in areas where fixed-route operations are difficult to sustain, but many-to-many door-to-door service can create long vehicle tours, low vehicle utilization, and fragile schedules under dynamic demand. Meeting-point service designs address this operational problem by asking passengers to walk to designated pickup or dropoff locations so that vehicles can consolidate routes. The same design choice also creates a passenger-response problem: offers that reduce vehicle travel may be rejected when walking, waiting, in-vehicle time, or fare burdens are too high. For logistics and operations management, the key question is therefore not whether meeting points are always preferable, but how bidirectional pickup and dropoff consolidation changes routing intensity, served share, and rejection patterns under tested service-design conditions.

This paper studies a many-to-many DRT setting with bidirectional meeting-point sets. For each request r, the operator constructs candidate pickup meeting points M_r^P near the origin and candidate dropoff meeting points M_r^D near the destination. The system then evaluates feasible service bundles, selects a routing-feasible offer, presents a single offer to the passenger, and samples acceptance through a binary-logit response model. Accepted requests enter a rolling-horizon dispatch process that periodically reoptimizes active vehicle routes. This passenger-response-aware simulation framework treats passenger response as an evaluation mechanism for generated offers, not as an empirically calibrated behavioral validation or an endogenous routing control.

The study is motivated by a gap between meeting-point DARP research and online DRT operations. Prior meeting-point formulations have demonstrated the operational value of walking flexibility, but they often focus on pickup-side consolidation or assume that assigned meeting points are accepted. Dynamic DRT and rolling-horizon routing studies, meanwhile, represent online fleet operations but generally retain door-to-door service or deterministic request acceptance. Combining bidirectional meeting points, single-offer binary-logit response, and rolling-horizon routing allows the paper to examine both sides of the service-design trade-off: route consolidation among served trips and demand loss through choice rejection or feasibility rejection.

The first contribution is a service-design formulation for many-to-many DRT with bidirectional pickup and dropoff meeting points. The formulation separates candidate meeting-point generation, feasible offer construction, passenger response, and dynamic dispatch so that each design layer can be inspected against operational metrics. This structure supports denominator-aware comparisons such as vehicle-km per served trip, vehicle-km per original request, served share, choice rejection, and feasibility rejection.

The second contribution is a passenger-response-aware simulation mechanism for evaluating bidirectional meeting-point offers. The binary-logit layer represents heterogeneous simulated passenger types and an outside option under a single-offer mechanism. It does not claim empirical population calibration; instead, it makes explicit how walking, waiting, in-vehicle time, and fare enter the acceptance evaluation and how accepted requests become the realized input to rolling-horizon operations.

The third contribution is an evidence-bounded operational trade-offs narrative for meeting-point DRT. Formal Phase 6 evidence is used to assess whether bidirectional meeting-point consolidation can reduce routing intensity among served trips under tested synthetic service-design conditions, while tracking coverage loss, passenger-response trade-offs, and rejection decomposition. Matched-coverage controls, fixed-accepted-set decompositions, robustness packages, passenger-type heterogeneity, post-hoc welfare accounting, and algorithm diagnostics are treated as diagnostic or supporting evidence rather than headline claims.

The remainder of this paper is organized as follows. Section 2 positions the study within DARP, meeting-point DRT, passenger choice, and dynamic DRT operations. Section 3 defines the service-design model and binary-logit response mechanism. Section 6 describes the rolling-horizon heuristic and the simplified ex-post routing diagnostic. Section 7 reports the evidence layers and denominator-aware experimental narrative. Section 8 discusses managerial and operational implications, and Section 9 summarizes conditional findings, limitations, and future work.

2. Literature Review

2.1. DARP, DRT, and On-Demand Mobility Operations

The Dial-a-Ride Problem (DARP) models passenger transport with origin–destination requests, time windows, ride-time limits, vehicle capacities, and fleet routing decisions [1, 2]. It has long provided an operations-research foundation for paratransit and shared passenger transport, and more recent work connects DARP methods to demand-responsive transit (DRT), ridepooling, and on-demand mobility logistics [3, 4]. In these systems, the operator must manage a fleet under sparse and dynamic request patterns, making routing intensity, service coverage, and operational reliability central logistics and operations concerns.

Classical DARP assumes door-to-door service. This maximizes passenger convenience but can fragment vehicle tours when many-to-many requests are spatially dispersed. Meeting points or virtual stops introduce service consolidation into the passenger-transport setting: passengers walk a bounded distance so that vehicles can serve more compatible stops and reduce detours. This paper treats meeting-point DRT as an operational service-design problem, not only as a routing variant, because the consolidation choice changes both fleet operations and passenger willingness to accept the offered service.

2.2. Meeting-Point DRT and Service Consolidation

Meeting points were introduced into shared-ride and ridepooling research as a way to trade small walking burdens for route consolidation [5, 6]. In the DARP literature, Cortenbach et al. [7] formalize the DARP with meeting points (DARPmp), allowing pickup-side flexibility while retaining fixed dropoff destinations. That work demonstrates the operational value of flexible pickup locations, but its single-sided design leaves dropoff-side consolidation outside the model and assumes that assigned passengers comply with the service offer.

Bidirectional walking flexibility has also been studied in ridepooling contexts. Fielbaum et al. [8] show that allowing both pickup and dropoff walking can reduce vehicle detours in static shared-ride settings. The present paper transfers that service-consolidation idea to many-to-many DRT operations with online request arrivals, rolling-horizon routing, and simulated passenger choice. The bidirectional design is therefore positioned as a logistics and operations service-design lever: it may lower routing intensity for served trips, but it can also alter served share and rejection composition.

2.3. Passenger Choice in Service Design

Discrete choice models provide a standard framework for representing heterogeneous passenger preferences over service attributes such as walking, waiting, in-vehicle time, and price [9, 10, 11, 12]. Most DRT routing models represent rejection as a feasibility outcome rather than an acceptance response: if a request can be inserted within the constraints, it is treated as served. That deterministic treatment is limiting for meeting-point service designs because walking burden and service delay are precisely the attributes that may cause passengers to reject an offer.

This paper uses a single-offer binary-logit mechanism as a simulation construct. The operator presents one routing-feasible bundle, and the simulated passenger either accepts that bundle or takes an outside option. Passenger types are used to represent a range of service sensitivities within the simulation; they are not empirical population segments validated from field data. This choice layer makes it possible to distinguish choice rejection from feasibility rejection and to evaluate passenger-segment monitoring implications without claiming field equity validation.

2.4. Dynamic DRT Routing and Rolling-Horizon Operations

DRT is commonly operated online: requests arrive over time, vehicles move while new offers are being made, and dispatch plans must be updated repeatedly. Dynamic vehicle routing and online DARP research therefore use insertion heuristics, metaheuristics, and rolling-horizon approaches to keep decisions tractable under uncertainty [13, 14, 3]. Rolling-horizon methods solve a finite active window, commit near-term decisions, and then reoptimize as new information arrives [15]. This structure is well suited to DRT operations because it represents the practical rhythm of dispatch rather than a clairvoyant offline plan.

Adaptive Large Neighborhood Search (ALNS) is widely used for DARP and related routing problems because destroy-and-repair operators can search large neighborhoods while respecting feasibility constraints [16]. Wu et al. [17] apply rolling-horizon ALNS to many-to-many DRT, showing the operational importance of periodic reoptimization. Their setting remains door-to-door and deterministic with respect to passenger acceptance. The present paper combines rolling-horizon operations with bidirectional meeting-point service consolidation and binary-logit passenger response.

2.5. Research Gap and Positioning

Table 1 summarizes the positioning of this paper relative to the most closely related prior works. The contribution is a passenger-response-aware simulation framework for bidirectional meeting-point DRT service design: bidirectional pickup and dropoff meeting points, single-offer binary-logit response, and rolling-horizon many-to-many operations are evaluated together while keeping diagnostic evidence roles separate from the primary behavioral narrative.

3. Problem Formulation

3.1. Network and Meeting Points

We model the service area as a directed graph $G = (N, A)$, where N is the set of nodes comprising passenger origins, destinations, and meeting points, and A is the set of directed arcs. Travel time between any two locations $i, j \in$

Table 1: Positioning relative to prior work

Study	Bidirectional MPs	Passenger Choice	Rolling Horizon	Many-to-M
Cortenbach et al. (2024)	No (single-sided)	No	No	Yes
Wu et al. (2025)	No	No	Yes	Yes
Fielbaum et al. (2021)	Yes (ridepooling)	No	No	Partia
Alonso-Mora et al. (2017)	No	No	No	Yes
This paper	Yes	Yes (binary logit)	Yes	Yes

N is given by $t(i,j) = d(i,j)/\bar{v}$, where $d(i,j)$ denotes Euclidean distance and $\bar{v}$ is the average vehicle speed.

A finite set $\mathcal{M}$ of pre-defined fixed meeting points (e.g., bus stops, commercial landmarks) is given. For each request r, the candidate pickup set $M_r^P \subseteq \mathcal{M}$ is the subset of meeting points within walking radius ρ^P of the passenger's origin o_r, and the candidate dropoff set $M_r^D \subseteq \mathcal{M}$ is the subset within walking radius ρ^D of the destination δ_r. Formally:

$$M_r^P = \big\{m \in \mathcal{M} \mid d(o_r,\, m) \le \rho^P\big\}, \qquad M_r^D = \big\{m \in \mathcal{M} \mid d(\delta_r,\, m) \le \rho^D\big\}. \tag{1}$$

Each request $r \in \mathcal{R}$ is characterized by a tuple $(o_r, \delta_r, e_r, l_r, k_r)$, where $[e_r, l_r]$ is the pickup time window and $k_r \in \mathcal{K}$ denotes the passenger type. Requests arrive online over a continuous planning horizon. The fleet $\mathcal{V}$ consists of vehicles with heterogeneous capacities Q_v and maximum shift durations $T_v^{\max}$.

3.2. Service Offer Bundle

For each request r, a *service offer bundle* is a tuple

$$b = (m_r^P,\ m_r^D,\ \tau_r,\ p_r),$$

where:

- $m_r^P \in M_r^P$ is the assigned pickup meeting point,
- $m_r^D \in M_r^D$ is the assigned dropoff meeting point,
- $\tau_r \in [e_r, l_r]$ is the scheduled pickup time at m_r^P,
- $p_r \geq 0$ is the fare charged to the passenger (fixed in this model).

The set of all feasible bundles for request r is $\mathcal{B}_r = M_r^P \times M_r^D$ (with τ_r and p_r treated as service parameters determined by the system). The service offer bundle b is the central decision object: the system selects one bundle from $\mathcal{B}_r$ or rejects the request.

The bidirectional meeting-point structure distinguishes this model from prior work. In many-to-one DRT, only the pickup side is flexible; here pickup and dropoff meeting points are both decision attributes, creating a richer trade-off between vehicle routing efficiency and passenger walking burden.

3.3. Routing Cost and Welfare Accounting

For offer generation and route construction, the system scores feasible insertions using a weighted sum of operational and passenger service costs:

$$\min \quad \alpha_1 C^{op} + \alpha_2 C^{wait} + \alpha_3 C^{walk} + \alpha_4 C^{IVT}, \tag{2}$$

where:

$$C^{op} = \sum_{v \in \mathcal{V}} c_v \cdot D_v, \tag{3}$$

$$C^{wait} = \sum_{r \in \mathcal{R}^{\text{acc}}} (\tau_r - t_r^{\text{req}}), \tag{4}$$

$$C^{walk} = \sum_{r \in \mathcal{R}^{\text{acc}}} \big(d(o_r, m_r^P) + d(m_r^D, \delta_r)\big), \tag{5}$$

$$C^{IVT} = \sum_{r \in \mathcal{R}^{\text{acc}}} \text{IVT}_{rb}. \tag{6}$$

Here D_v is the total distance driven by vehicle v, c_v is cost per unit distance, and t_r^{req} is the time request r was submitted. The weights $\alpha_1, \ldots, \alpha_4 \geq 0$ allow the operator to balance vehicle efficiency against service quality.

For post-hoc welfare accounting, we define a social welfare metric that includes a rejection penalty $\Gamma > 0$:

$$W = \sum_{r \in \mathcal{R}} \big[z_r \cdot U_{rb^*} - (1 - z_r) \cdot \Gamma\big]. \tag{7}$$

Γ enters only the ex-post welfare accounting discussed in Section 7; it does *not* affect routing, offer generation, or acceptance. In particular, it does not enter the routing objective (Equation 2), the bundle selection rule, or the binary-logit acceptance probability. The separation keeps operational decisions fixed while sensitivity accounting varies Γ after the simulation.

3.4. Operational Feasibility Conditions

The routing layer enforces the following feasibility conditions for requests whose offers are accepted. These conditions define the simulation's feasible routing state; they are not presented as a complete arc-flow mixed-integer programming formulation.

Vehicle Capacity..

$$q_v(i) \leq Q_v \quad \forall v \in \mathcal{V},\ \forall i \in \sigma_v \tag{8}$$

Time Windows..

$$s_{v_r,\, m_r^P} \geq e_r \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{9}$$

$$s_{v_r,\, m_r^P} \leq l_r \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{10}$$

In-Vehicle Ride Time..

$$s_{v_r,\, m_r^D} - s_{v_r,\, m_r^P} \leq L^{\max} \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{11}$$

Walking Radius..

$$d(o_r,\, m_r^P) \leq \rho^P \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{12}$$

$$d(\delta_r,\, m_r^D) \leq \rho^D \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{13}$$

Precedence: Pickup Before Dropoff..

$$\pi_r^P < \pi_r^D \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{14}$$

$$s_{v_r,\, m_r^P} + t(m_r^P,\, m_r^D) \leq s_{v_r,\, m_r^D} \quad \forall r \in \mathcal{R}^{\mathrm{acc}} \tag{15}$$

Route Time Consistency..

$$s_{v,\, i+1} \geq s_{v,\, i} + t(\sigma_v(i),\, \sigma_v(i+1)) \quad \forall v \in \mathcal{V},\ \forall i < |\sigma_v| \tag{16}$$

Maximum Route Duration..

$$s_{v,\, |\sigma_v|} - s_{v,\, 1} \leq T_v^{\max} \quad \forall v \in \mathcal{V} \tag{17}$$

Domain Constraints..

$$v_r \in \mathcal{V} \qquad \forall r \in \mathcal{R}^{\text{acc}} \tag{18}$$

$$m_r^P \in M_r^P \qquad \forall r \in \mathcal{R}^{\text{acc}} \tag{19}$$

$$m_r^D \in M_r^D \qquad \forall r \in \mathcal{R}^{\text{acc}} \tag{20}$$

Table 2 summarizes the operational feasibility conditions.

Table 2: Summary of operational constraints

Label	**Class**	**Variables Involved**
8	Vehicle Capacity	$q_v(i)$, Q_v, σ_v
9	Time Windows (early)	s_{v_r,m_r^P}, e_r
10	Time Windows (late)	s_{v_r,m_r^P}, l_r
11	In-Vehicle Ride Time	s_{v_r,m_r^D}, s_{v_r,m_r^P}, $L^{\max}$
12	Walking Radius (pickup)	$d(o_r, m_r^P)$, ρ^P
13	Walking Radius (dropoff)	$d(\delta_r, m_r^D)$, ρ^D
14	Precedence (position)	π_r^P, π_r^D
15	Precedence (time)	s_{v_r,m_r^P}, s_{v_r,m_r^D}, $t(m_r^P, m_r^D)$
16	Route Time Consistency	$s_{v,i}$, $s_{v,i+1}$, $t(\sigma_v(i), \sigma_v(i+1))$
17	Maximum Route Duration	$s_{v,1}$, $s_{v,\lvert\sigma_v\rvert}$, $T_v^{\max}$
18	Domain (vehicle)	v_r
19	Domain (pickup point)	m_r^P, M_r^P
20	Domain (dropoff point)	m_r^D, M_r^D

3.5. Online Offer and Response Variables

Upon arrival of request r, the system immediately determines the *online offer/routing decision*:

$$\mathbf{a}_r = (m_r^P,\ m_r^D,\ v_r,\ \pi_r^P,\ \pi_r^D,\ \tau_r), \tag{21}$$

where $m_r^P \in M_r^P$ and $m_r^D \in M_r^D$ are the assigned meeting points, $v_r \in \mathcal{V}$ is the assigned vehicle, $\pi_r^P < \pi_r^D$ are the insertion positions of the pickup and dropoff nodes in vehicle v_r's route, and τ_r is the scheduled pickup time implied by the insertion. The offer decision must be made online — before future requests are known — and subject to all feasibility constraints above. The binary variable z_r is not optimized by the system; it is the realized passenger response generated after the offer is presented, as defined in Section 4.

4. Passenger Choice Model

4.1. Binary Logit Utility Function

We model passenger response using a two-alternative binary logit model. For request r of passenger type $k \in \mathcal{K}$, the deterministic utility of service offer bundle $b = (m_r^P, m_r^D, \tau_r, p_r)$ is:

$$U_{rb}^k = \beta_1^k \cdot \mathrm{Walk}_{rb} + \beta_2^k \cdot \mathrm{Wait}_{rb} + \beta_3^k \cdot \mathrm{IVT}_{rb} + \beta_4^k \cdot p_r \tag{22}$$

where:

- $\mathrm{Walk}_{rb} = d(o_r, m_r^P) + d(m_r^D, \delta_r)$ is the total walking distance (pickup walk plus dropoff walk) in meters,

- $\text{Wait}_{rb} = \tau_r - t_r^{\text{req}}$ is the waiting time from request submission to scheduled pickup (minutes),
- $\text{IVT}_{rb} = s_{v_r, m_r^D} - s_{v_r, m_r^P}$ is the in-vehicle travel time from pickup to dropoff meeting point (minutes),
- $p_r \geq 0$ is the fare (CNY),
- $\beta_1^k, \beta_2^k, \beta_3^k \leq 0$ are disutility coefficients for walking, waiting, and in-vehicle time (negative: more time/distance reduces utility),
- $\beta_4^k \leq 0$ is the price disutility coefficient.

The random utility is $U_{rb}^k + \varepsilon_{rb}$ where ε_{rb} is i.i.d. Gumbel$(0, 1)$, yielding the standard random-utility form. The utility function captures the bidirectional walking cost explicitly through Walk_{rb}, which depends on both m_r^P and m_r^D — a key distinction from one-sided meeting-point models.

4.2. Single-Offer Mechanism and Binary Logit Acceptance

The system operates a *single-offer mechanism*: Layer 1 selects the best feasible bundle $b^* \in \mathcal{B}_r$ and presents *only* b^* to the passenger. The passenger never observes the full set $\mathcal{B}_r$; they choose between accepting b^* or taking the outside option (e.g., private car, walking, trip cancellation).

The utility of the outside option is normalized to zero:

$$U_{r0}^k = 0 \tag{23}$$

Because the passenger's choice set contains exactly two alternatives — b^* and the outside option — the acceptance probability follows a *binary logit* model:

$$P_{\text{accept}}(b^*) = \frac{\exp(U^k_{rb^*})}{\exp(U^k_{r0}) + \exp(U^k_{rb^*})} = \frac{\exp(U^k_{rb^*})}{1 + \exp(U^k_{rb^*})} \tag{24}$$

The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \tag{25}$$

This formulation is behaviorally consistent with the single-offer mechanism: the passenger responds to the one bundle they are shown, not to a hypothetical full menu. The standard random-utility interpretation (i.i.d. Gumbel$(0,1)$ errors) is preserved for this two-alternative setting.

4.3. Passenger Heterogeneity

We distinguish three simulation-range passenger types $\mathcal{K} = \{\text{price}, \text{time}, \text{walk}\}$ with different sensitivity profiles. Table 3 summarizes the qualitative ordering of coefficients. These types are simulation constructs for monitoring service-design heterogeneity; they are not empirical demographic groups or field equity validation.

Table 3: Passenger type coefficient profiles (qualitative ordering)

Type	β^k_1 (Walk)	β^k_2 (Wait)	β^k_3 (IVT)	β^k_4 (P
Price-sensitive	moderate	moderate	moderate	high (most
Time-sensitive	moderate	high (most negative)	high (most negative)	lo
Walk-sensitive	high (most negative)	moderate	moderate	mode

Formally, the ordering constraints are:

$$|\beta_4^{\text{price}}| > |\beta_4^{\text{time}}|, \quad |\beta_4^{\text{price}}| > |\beta_4^{\text{walk}}| \tag{26}$$

$$|\beta_2^{\text{time}}| + |\beta_3^{\text{time}}| > |\beta_2^{\text{price}}| + |\beta_3^{\text{price}}| \tag{27}$$

$$|\beta_1^{\text{walk}}| > |\beta_1^{\text{price}}|, \quad |\beta_1^{\text{walk}}| > |\beta_1^{\text{time}}| \tag{28}$$

Baseline parameter values for simulation:

$$(\beta_1^{\text{price}},\ \beta_2^{\text{price}},\ \beta_3^{\text{price}},\ \beta_4^{\text{price}}) = (-0.005,\ -0.04,\ -0.02,\ -0.15) \tag{29}$$

$$(\beta_1^{\text{time}},\ \beta_2^{\text{time}},\ \beta_3^{\text{time}},\ \beta_4^{\text{time}}) = (-0.005,\ -0.10,\ -0.08,\ -0.03) \tag{30}$$

$$(\beta_1^{\text{walk}},\ \beta_2^{\text{walk}},\ \beta_3^{\text{walk}},\ \beta_4^{\text{walk}}) = (-0.020,\ -0.04,\ -0.02,\ -0.05) \tag{31}$$

Units: β_1 in utility/meter, β_2 and β_3 in utility/minute, β_4 in utility/CNY. The heterogeneous type structure allows the simulation to monitor how service-design burdens differ across sensitivity profiles. It should not be read as a validated demographic equity model.[1]

[1]**Parameter plausibility.** The implied value of time (VOT) for the walk-sensitive type — the most policy-relevant comparator for bidirectional DRT users — is $\text{VOT}_{\text{walk}} = (|\beta_1^{\text{walk}}|/|\beta_4^{\text{walk}}|) \times v_w = (0.020/0.05) \times 80 = 32.0\,\text{CNY/min}$, $\text{VOT}_{\text{wait}} = |\beta_2^{\text{walk}}|/|\beta_4^{\text{walk}}| = 0.04/0.05 = 0.80\,\text{CNY/min}$, and $\text{VOT}_{\text{IVT}} = 0.02/0.05 = 0.40\,\text{CNY/min}$ (walking speed $v_w = 80\,\text{m/min}$ assumed). The wait and IVT VOT values (0.80 and 0.40 CNY/min) fall within the Chinese urban reference range of 0.8–1.2 CNY/min (waiting) and 0.2–0.4 CNY/min (IVT) reported by Shao et al. [18] for commuters and Li et al. [19] for DRT users. The walk VOT (32.0 CNY/min) exceeds the literature range (0.4–0.6 CNY/min); this reflects the strong walking disutility assumed for the walk-sensitive type ($|\beta_1^{\text{walk}}| = 0.020\,\text{utils/meter}$) combined with low price sensitivity ($|\beta_4^{\text{walk}}| = 0.05\,\text{utils/CNY}$). The resulting ratio ($0.020/0.05 \times 80 = 32.0\,\text{CNY/min}$) is an upper bound; calibration against revealed-preference data from DRT users would likely yield lower walking disutility coeffi-

4.4. Scenario Semantics

The scenario labels used later in the manuscript are simulation labels. The baseline synthetic grid is a controlled service-design setting, and the Beijing-inspired synthetic grid is a semi-realistic synthetic grid used for robustness-oriented sensitivity. Neither scenario constitutes field validation in Beijing, public-data ingestion, or empirical calibration.

Worked utility example.. Consider a walk-sensitive passenger (k = walk) with $\beta_1^{\text{walk}} = -0.020\,\text{utils/m}$, $\beta_2^{\text{walk}} = -0.04\,\text{utils/min}$, $\beta_3^{\text{walk}} = -0.02\,\text{utils/min}$, and $\beta_4^{\text{walk}} = -0.05\,\text{utils/CNY}$. Suppose the system offers a bundle with walking distance $d_{\text{walk}} = 300\,\text{m}$, waiting time $t_{\text{wait}} = 5\,\text{min}$, in-vehicle time $t_{\text{ivt}} = 20\,\text{min}$, and fare $p = 8\,\text{CNY}$. The deterministic utility is:

$$\begin{aligned} U &= \beta_1 \cdot d_{\text{walk}} + \beta_2 \cdot t_{\text{wait}} + \beta_3 \cdot t_{\text{ivt}} + \beta_4 \cdot p \\ &= (-0.020)(300) + (-0.04)(5) + (-0.02)(20) + (-0.05)(8) \\ &= -6.0 - 0.2 - 0.4 - 0.4 = -7.0\ \text{utils}. \end{aligned}$$

With the outside option utility normalized to zero ($U_{r0}^k = 0$, Equation 23), the acceptance probability is:

$$P_{\text{accept}} = \frac{e^{-7.0}}{e^{-7.0} + e^{0}} = \frac{0.000912}{0.000912 + 1} \approx 0.00091. \tag{32}$$

This near-zero acceptance probability (0.091%) illustrates why the offer quality generated by the online algorithm is critical: under the $U_0 = 0$ normalization, any bundle with strongly negative utility will be rejected with very

cients. This limitation is discussed in Section 8 (VOT interpretation) and Section 9. type's defining characteristic. Further VOT interpretation is treated as calibration-dependent in the implications section.

high probability. The system must therefore offer bundles with substantially higher utility (e.g., by reducing walking distance, lowering wait times, or offering price discounts) to achieve meaningful acceptance rates. Reducing walking to 100 m would raise U to -3.0, yielding $P_{\text{accept}} \approx 0.047$ (4.7% acceptance), illustrating the sensitivity of demand to meeting point placement.

5. Three-Layer Coupled Model

The many-to-many DRT system with bidirectional meeting points operates through three sequentially coupled layers, executed online upon each request arrival and periodically re-optimized via a rolling horizon mechanism. Figure 2 illustrates the architecture. The three layers are: (1) service offer generation, (2) passenger response, and (3) dynamic dispatch and re-optimization.

5.1. Layer 1: Service Offer Generation

Upon arrival of request r at time t_r^{req}, the system generates a set of candidate service offer bundles $\hat{\mathcal{B}}_r \subseteq \mathcal{B}_r$ as follows:

1. **Candidate pickup set:** Compute $M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to o_r.
2. **Candidate dropoff set:** Compute $M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to δ_r.
3. **Bundle enumeration:** For each pair $(m^P, m^D) \in M_r^P \times M_r^D$, evaluate feasibility against constraints 8–16 for each vehicle $v \in \mathcal{V}$ and each insertion position pair (π^P, π^D).

4. **Bundle selection:** Select the bundle $b^* = \arg\min_{b \in \hat{\mathcal{B}}_r} \Delta C_{\text{routing}}(b, \{\sigma_v\})$ that minimizes the deterministic incremental routing cost $\alpha_1 \Delta C^{op} + \alpha_2 \Delta C^{wait} + \alpha_3 \Delta C^{walk} + \alpha_4 \Delta C^{IVT}$ (see Section 6 for implementation details).

The output of Layer 1 is the online offer/routing decision $\mathbf{a}_r$ (Equation 21) and the service offer bundle b^* presented to the passenger.

5.2. Layer 2: Passenger Response

Given the offered bundle b^*, passenger r of type k accepts with probability $P_{\text{accept}}(b^*)$ (Equation 24) and rejects with probability $1 - P_{\text{accept}}(b^*)$. The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \tag{33}$$

If $z_r = 1$, the vehicle route σ_{v_r} is updated by inserting the pickup node m_r^P at position π_r^P and the dropoff node m_r^D at position π_r^D. If $z_r = 0$, the offer is rejected and vehicle routes remain unchanged.

The coupling between Layer 1 and Layer 2 follows a *route-then-sample* design (Section 6): Layer 1 selects b^* by minimizing deterministic incremental routing cost; the binary logit acceptance probability $P_{\text{accept}}(b^*)$ is computed after bundle selection and used only for the Bernoulli acceptance sampling in Layer 2. This decoupling avoids the perverse incentive where multiplying P_{accept} into the routing objective would favor low-acceptance (poor service) offers. Integrating P_{accept} into a welfare-consistent anticipatory objective remains a direction for future work.

5.3. Layer 3: Dynamic Dispatch and Rolling Horizon Re-optimization

Vehicle dispatch follows the committed route schedules $\{\sigma_v, \mathbf{s}_v\}_{v\in\mathcal{V}}$. Every Δ minutes, a rolling horizon re-optimization is triggered over the active request window $[t^{\text{now}},\ t^{\text{now}} + H]$:

1. **Active set:** Identify $\mathcal{R}^{\text{act}}(t^{\text{now}})$ — all accepted requests not yet completed at time t^{now}.
2. **Re-optimization:** Apply ALNS (Adaptive Large Neighborhood Search) to reassign and reorder requests in $\mathcal{R}^{\text{act}}(t^{\text{now}})$ across vehicles, subject to constraints 8–16.
3. **Commitment:** Commit the updated routes for the next Δ minutes; nodes already served are locked and cannot be reassigned.

The rolling horizon creates a feedback loop from Layer 3 back to Layer 1: updated vehicle schedules after re-optimization change the feasibility and cost of future bundle insertions, influencing the next Layer 1 evaluation.

$$\text{Re-optimize at times } t^{\text{now}} = 0,\ \Delta,\ 2\Delta,\ 3\Delta,\ \ldots \tag{34}$$

Typical parameter values: $H = 30$ minutes (horizon window), $\Delta = 5$ minutes (re-optimization frequency). These are configurable and subject to sensitivity analysis in the sensitivity analysis (Section 7).

5.4. Layer Coupling Summary

Table 4 summarizes the information flows between layers.

Table 5 details the event sequence and information flows within a single planning cycle; in particular, it shows that Bernoulli sampling ($z_r \sim$

Table 4: Three-layer model coupling structure

Layer	Input	Output	Coupling
1: Service Generation	Request r, routes $\{\sigma_v\}$	Bundle b^*, decision $\mathbf{a}_r$	Feeds Layer 2
2: Passenger Response	Bundle b^*, type k	Acceptance z_r	Updates routes if $z_r =$
3: Rolling Horizon	Active set $\mathcal{R}^{\text{act}}$	Updated routes $\{\sigma_v\}$	Feeds back to Layer 1

Bernoulli($P_{\text{accept}}(b^*)$)) occurs exclusively in Layer 2, after bundle b^* has been selected by Layer 1.

The three-layer architecture captures the essential feedback structure of online DRT operations: routing decisions shape the offers available to passengers, passenger choices determine realized demand, and periodic re-optimization adapts routes to the evolving state. The notation table (Table A.7) in the appendix provides a complete reference for all symbols used in this paper.

6. Solution Methodology

The problem defined in Section 3 is NP-hard (see Section 6.3). We develop two complementary solution approaches: an MILP-based simplified ex-post diagnostic solved by Gurobi for small instances, and a rolling horizon Adaptive Large Neighborhood Search (ALNS) heuristic for real-time operation at scale.

6.1. Simplified Ex-Post Diagnostic: MILP Formulation

For small instances, we formulate the static batch routing problem as a Mixed-Integer Linear Program (MILP) to compute the optimal routing cost

Table 5: Timing and decision sequence in the rolling horizon three-layer model. Each row represents one event within a single planning cycle. Re-optimization triggers fire at $t^{\text{now}} = 0, \Delta, 2\Delta, \dots$ (Equation 34).

Event	**Layer**	**Variables / outcomes**	**Information Flow**	**Bernoulli?**
Request r arrives at t_r^{req}	1	$m_r^P,\ m_r^D,\ v_r,\ \pi_r^P,\ \pi_r^D$	Routes $\{\sigma_v\}$ in; bundle b^* out	No
Bundle b^* presented to passenger	2	z_r	b^* in; acceptance z_r out	Yes: $z_r \sim \text{Bern}(P_{\text{acc}})$
If $z_r = 1$: route σ_{v_r} updated	3	(route update)	z_r, b^* in; updated σ_{v_r} out	No
Re-opt trigger at $t = k\Delta$	3	$\{\sigma_v\}$ (all vehicles)	$\mathcal{R}^{\text{act}}$ in; optimised $\{\sigma_v\}$ out	No
Updated routes fed back for next request	1	—	Updated $\{\sigma_v\}$ available to Layer 1	No

for a given accepted set. The MILP serves as a *simplified ex-post diagnostic over fixed accepted sets*, not a validation tool for the complete simulation framework, which includes offer generation, routing, and stochastic passenger response.

Decision Variables.. The MILP uses binary assignment variables $x_{r,m^P,m^D,v} \in \{0,1\}$, equal to 1 if request r is assigned to meeting point pair $(m^P, m^D) \in M_r^P \times M_r^D$ and served by vehicle $v \in \mathcal{V}$. Continuous variables $s_{v,i} \geq 0$ represent the scheduled service time at node i in vehicle v's route. The MILP is solved only for the fixed realized set $\mathcal{R}^{\text{acc}}$; it does not optimize the passenger response variable z_r.

Objective.. The MILP minimizes the same weighted cost (Equation 2), with each cost component linearized in terms of $x_{r,m^P,m^D,v}$ and $s_{v,i}$.

Constraints.. The formulation enforces the operational feasibility conditions (8)–(20). Capacity constraint (8) is linearized using standard load-tracking variables. Time-window constraints (9)–(10), ride-time constraint (11), and route time consistency (16) are expressed as linear inequalities in $s_{v,i}$. Walking radius constraints (12)–(13) are enforced implicitly by restricting the domain of $x_{r,m^P,m^D,v}$ to feasible meeting point pairs.

Solver and Scope.. The MILP is solved using Gurobi 10 with a 300-second time limit. Any reported solver gap is interpreted only within this simplified fixed-accepted-set diagnostic.

Simplified ex-post diagnostic role. The MILP serves as a *simplified ex-post diagnostic over fixed accepted sets*, not a real-time solver or a complete dynamic routing comparison. Given the realized accepted set $\mathcal{R}^{\text{acc}}$ produced by a completed ALNS run — i.e., the set of requests for which the sampled acceptance indicator $z_r = 1$ — the MILP solves the optimal vehicle routing for that *fixed* accepted set. Formally, the MILP treats z_r as a known constant (equal to 1 for $r \in \mathcal{R}^{\text{acc}}$ and 0 otherwise) and optimizes only over the assignment variables $x_{r,m^P,m^D,v}$ and schedule variables $s_{v,i}$. It does *not* re-optimize the acceptance decisions: the stochastic passenger responses are realized outcomes of the ALNS run, not MILP decision variables. The resulting MILP objective value is therefore a fixed-set diagnostic comparator, and the diagnostic gap $\delta = (C^{\text{ALNS}} - C^{\text{MILP}})/C^{\text{MILP}} \times 100\%$ describes the difference between the heuristic route and the simplified fixed-set diagnostic solution only. Instances with up to 30–50 passengers and 5–8 vehicles are tractable within the time limit.

6.2. Heuristic Algorithm: Rolling Horizon ALNS

For real-time operation, we develop a rolling horizon ALNS heuristic that processes arriving requests online and periodically re-optimizes committed routes. The algorithm follows the three-layer architecture of Section 5.

Online Insertion Mechanism.. The ALNS heuristic uses a *route-then-sample* mechanism that decouples stochastic passenger acceptance from deterministic routing optimization. At each request arrival, the system performs the following steps:

1. **Bundle selection (Layer 1):** The system selects the best feasible bundle b^* by minimizing the *deterministic* incremental routing cost $\Delta C_{\text{routing}}(b, \{\sigma_v\})$, evaluated as the weighted sum $\alpha_1 \Delta C^{op} + \alpha_2 \Delta C^{wait} + \alpha_3 \Delta C^{walk} + \alpha_4 \Delta C^{IVT}$ of inserting bundle b into the current vehicle schedules $\{\sigma_v\}$. The acceptance probability $P_{\text{accept}}(b^*)$ is *not* part of the bundle-selection objective.
2. **Acceptance sampling (Layer 2):** After bundle selection, the binary logit acceptance probability $P_{\text{accept}}(b^*) = \exp(U^k_{rb^*})/(1+\exp(U^k_{rb^*}))$ (Equation 24) is computed, and the passenger's response is sampled as $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$. If $z_r = 1$, the bundle is inserted into the vehicle route.

Rationale for route-then-sample.. This design separates the stochastic acceptance decision from the deterministic cost minimization. An alternative approach that integrates P_{accept} directly into the routing objective as

$\mathbb{E}[\Delta C] = P_{\text{accept}}(b) \times \Delta C_{\text{routing}}(b)$ would create a perverse incentive: bundles with lower acceptance probability (worse service quality) would appear to have lower expected routing cost, driving the system toward offering unattractive bundles. The route-then-sample approach avoids this pathology while preserving the structural benefit of passenger self-selection: passengers who would face high walking or waiting disutility reject the service, and the system concentrates vehicle resources on spatially compatible requests. Integrating P_{accept} into the routing objective in a welfare-consistent manner remains a direction for future work.

Rejection penalty (post-hoc only).. The rejection penalty Γ enters only the social welfare metric $W = \sum_r [z_r U_{rb^*} - (1 - z_r)\Gamma]$ (Section 7), which is a post-hoc evaluation tool. Γ does *not* enter the bundle selection, the routing objective, or the binary logit acceptance probability. The acceptance probability $P_{\text{accept}}(b^*)$ is invariant to Γ. This separation ensures that the operational decisions are consistent across different post-hoc welfare-accounting settings encoded through Γ.

6.2.1. Candidate Generation

Upon arrival of request r, the system generates candidate meeting point sets M_r^P and M_r^D by filtering $\mathcal{M}$ against walking radii ρ^P and ρ^D (constraints 12–13), then retaining the top-k^{top} candidates ranked by proximity to o_r and δ_r respectively. This yields $|M_r^P| \leq k^{\text{top}}$ pickup candidates and $|M_r^D| \leq k^{\text{top}}$ dropoff candidates. The filtering step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$ per request using a pre-sorted distance index.

6.2.2. Online Insertion Evaluation

For each bundle $(m^P, m^D) \in M_r^P \times M_r^D$, the algorithm enumerates all vehicles $v \in \mathcal{V}$ and all insertion position pairs (π^P, π^D) with $\pi^D > \pi^P$ (constraint 14). Feasibility against constraints (8), (9)–(10), (11), and (16) is checked in $O(1)$ per position pair using forward and backward schedule arrays maintained for each vehicle route. The best feasible insertion is selected by minimum incremental cost ΔC to the objective (2). After selecting b^*, the binary logit acceptance probability $P_{\text{accept}}(b^*) = \exp(U^k_{rb^*})/(1+\exp(U^k_{rb^*}))$ (Equation 24) is computed and the passenger's response is sampled as $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$.

6.2.3. ALNS Destroy and Repair Operators

Every Δ minutes, the rolling horizon re-optimization applies ALNS [16] to the active request set $\mathcal{R}^{\text{act}}(t^{\text{now}})$. The algorithm uses five operators:

1. **Request reassignment:** Remove k randomly selected requests from their current vehicle routes and reinsert them greedily into the best feasible positions across all vehicles.
2. **Pickup/dropoff pair swap:** Exchange the meeting point pair (m_r^P, m_r^D) assignments between two requests r and r', updating insertion positions accordingly.
3. **Insertion position exchange:** Swap the route positions of two service nodes on the same vehicle, subject to precedence and feasibility.
4. **Route segment reconstruction:** Remove a contiguous route segment from a vehicle, then reinsert the affected requests optimally across all vehicles.

5. **Worst removal:** Remove the k requests with the highest marginal cost contribution to the objective, then reinsert greedily.

Operator selection follows the standard ALNS roulette-wheel mechanism [16]: each operator maintains a score updated by its historical improvement contribution, and operators are selected with probability proportional to their score. Meeting point reassignment is integrated into the repair step of operators 1, 4, and 5: when a request is reinserted, the algorithm re-evaluates all (m^P, m^D) pairs in $M_r^P \times M_r^D$ and selects the pair that minimizes insertion cost.

6.2.4. Rolling Horizon Loop

The rolling horizon framework [15] triggers ALNS re-optimization at times $t^{\text{now}} = 0, \Delta, 2\Delta, \ldots$ (Equation 34). At each trigger, the algorithm operates on the active set $\mathcal{R}^{\text{act}}(t^{\text{now}})$ over the window $[t^{\text{now}},\ t^{\text{now}} + H]$ with $H = 30$ minutes and $\Delta = 5$ minutes. Committed nodes (already served or within the commitment horizon) are locked and cannot be reassigned.

Commitment assumption.. Once a service offer bundle $b_r^* = (m_r^P, m_r^D, \tau_r, p_r)$ is communicated to passenger r and accepted ($z_r = 1$), the assigned meeting points and scheduled pickup time are *committed*: they are not subject to re-optimization in subsequent rolling horizon cycles. This commitment is operationally necessary — passengers who have begun walking to a meeting point cannot be redirected — and is enforced by locking the corresponding nodes in the route graph before each ALNS re-optimization trigger. Formally, a node $i \in \sigma_v$ is committed if its scheduled service time satisfies $\tau_i \leq t^{\text{now}} + \delta_{\text{commit}}$, where $\delta_{\text{commit}} = 5$ minutes is the commitment horizon (equal to the re-optimization interval Δ). Uncommitted accepted nodes may be re-sequenced within the vehicle route to improve efficiency, but their assigned meeting points remain fixed.

The ALNS runs for 50 iterations per re-optimization trigger, with each iteration applying one destroy–repair operator pair. After re-optimization, updated routes are committed for the next Δ minutes and fed back to Layer 1 for subsequent request evaluations.

6.3. Computational Complexity

NP-Hardness.. The many-to-many DRT problem with bidirectional meeting points is NP-hard. This follows by reduction from the Dial-a-Ride Problem (DARP) [1]: for each request r, set the candidate pickup set to a singleton containing only the passenger's origin ($M_r^P = \{o_r\}$) and the candidate dropoff set to a singleton containing only the destination ($M_r^D = \{\delta_r\}$), with ρ^P and ρ^D set sufficiently large to include these singletons. The problem then reduces to standard DARP (door-to-door with no meeting point choice), which is NP-hard. The bidirectional meeting-point extension strictly generalizes this case by allowing $|M_r^P|, |M_r^D| \geq 1$, adding the joint optimization of m_r^P and m_r^D, so the problem is at least as hard.

Heuristic Complexity.. The per-request candidate generation step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$. Online insertion evaluation for a single request costs $O(|\mathcal{V}| \cdot |\sigma_v|^2 \cdot (k^{\text{top}})^2)$, where $|\sigma_v|$ is the average route length. Each ALNS iteration costs $O(n \cdot |\mathcal{V}| \cdot |\sigma_v|)$ where $n = |\mathcal{R}^{\text{act}}|$. The rolling horizon bounds the active set size: at any trigger, at most $H/\bar{\lambda}^{-1}$ requests are active, where $\bar{\lambda}$ is the average arrival rate. This keeps per-step complexity manageable even as the total number of requests grows.

Practical Performance.. On the 200-request, 15-vehicle simulation instances, the average decision time per request is below 1 second, satisfying the real-time requirement of online DRT operation. The simplified ex-post diagnostic over fixed accepted sets (Section 6.1) compares ALNS routing with the diagnostic MILP solution for a fixed accepted set on small instances. These diagnostic differences reflect the limited fixed-set setup, not an assessment of overall solution quality, since the MILP does not model stochastic passenger response.

7. Numerical Experiments

This section describes the evidence structure used to evaluate the bidirectional meeting-point DRT service design. The main result presentation is table-first: concrete values are reported in the formal Phase 6 table and figure, while the prose states the tested conditions, denominator meanings, and evidence roles. The formal evidence package is generated from `results/formal/phase06/`.

7.1. Evidence Roles and Denominators

The primary behavioral evidence comes from the formal Phase 6 main behavioral package. It compares service designs under the same experimental framework and supports conditional claims about routing intensity, served share, and rejection decomposition. The primary metrics are:

- vehicle-km per served trip, with vehicle kilometres divided by served trips;

- vehicle-km per original request, with vehicle kilometres divided by all requests;
- served share, with served requests divided by all requests;
- choice rejection, which records passengers who reject a feasible offer;
- feasibility rejection, which records requests for which no feasible offer is found.

These denominator labels are part of the claim boundary. Vehicle-km per served trip speaks to routing intensity among realized served trips, while vehicle-km per original request keeps coverage loss visible. Served share, choice rejection, and feasibility rejection are therefore reported with efficiency claims rather than treated as secondary details.

The formal Phase 6 package also contains evidence that is useful but not headline evidence. Matched-coverage outputs are a diagnostic coverage-confounding control. Fixed-accepted-set outputs are a diagnostic decomposition over fixed accepted sets. Robustness packages test sensitivity to service-design settings. Equity/type outputs support limited passenger-type monitoring implications. Gamma outputs are post-hoc welfare or sensitivity accounting. Algorithm outputs are an algorithm diagnostic, including a simplified ex-post diagnostic over fixed accepted sets. These roles are kept separate from the primary behavioral evidence.

7.2. Experimental Design

The experiments compare door-to-door service, single-sided meeting-point service, bidirectional service without passenger choice, the full passenger-response-aware bidirectional design, and targeted ablations. The scenario set includes a controlled synthetic grid and a Beijing-inspired synthetic grid. The Beijing-inspired synthetic grid is used as a robustness-oriented synthetic service-design setting, not as field validation or public-data evidence.

Passenger response is represented through the binary-logit single-offer mechanism defined in Section 4. For variants without passenger choice, inserted or served request counts are interpreted according to that variant's mechanism rather than as behavioral acceptance. The rolling-horizon routing layer is interpreted as the operational heuristic used to update active routes as requests arrive.

7.3. Primary Behavioral Evidence

The main narrative focuses on whether passenger-response-aware bidirectional meeting-point consolidation can lower routing intensity for served trips under tested synthetic service-design conditions. This claim is conditional. It must travel with served share, vehicle-km per original request, choice rejection, feasibility rejection, and passenger-type monitoring because meeting-point consolidation can make some offers less acceptable or infeasible.

Table 6 reports the main behavioral comparison from formal Phase 6 processed outputs. The table pairs routing-intensity metrics with served share, choice rejection, and feasibility rejection so that the denominator trade-off is visible. Figure 4 provides the corresponding visual summary of total vehicle-km and served share across the main service designs. Together, they support the qualitative interpretation that bidirectional pickup and dropoff consolidation can create lower vehicle-km per served trip for the requests that

Table 6: Formal Phase 6 main behavioral evidence for routing intensity and coverage

Method	Served share	Choice rejection	Feasibility rejection	Vehicle-km / served trip	Vehicle-km / original request
FullModel	15.8%	54.0%	30.3%	9.95	1.57
DoorToDoor	20.0%	56.3%	23.7%	14.67	2.90
SingleSidedPickup	19.9%	54.4%	25.6%	13.68	2.74
SingleSidedDropoff	19.4%	54.9%	25.8%	13.91	2.66

FullModel minus DoorToDoor paired differences across formal seeds and scales

Metric	Mean difference	95% bootstrap interval	Valid pairs
Served share	-4.3 pp	[-5.2, -3.4] pp	80
Choice rejection	-2.4 pp	[-4.9, 0.2] pp	80
Feasibility rejection	6.6 pp	[3.3, 9.9] pp	80
Vehicle-km per served trip	-4.71	[-5.24, -4.24]	80
Vehicle-km per original request	-1.33	[-1.48, -1.19]	80

Notes: Generated from f
Panel means use the form
paired bootstrap table. S
quests; vehicle-km per ser
original request divides v
rejection rates use origina
seed-scale pairs and are d
hypothesis-test significanc

accept feasible offers, while reducing coverage relative to some door-to-door or single-sided designs. The result is therefore an operational trade-off, not an unconditional superiority claim.

7.4. Diagnostic and Robustness Roadmap

The diagnostic evidence is reported as a roadmap rather than as the main estimate. The diagnostic coverage-confounding control asks whether lower vehicle-km per served trip remains visible when served share is brought closer across compared designs. The diagnostic decomposition over fixed accepted sets isolates routing and meeting-point effects after the accepted request set is fixed. Robustness checks examine how the patterns change under service-design settings such as walking tolerance, fleet stress, and utility sensitivity. Passenger-type outcomes are used only as limited passenger-segment monitoring evidence, not as empirical population-equity validation.

Gamma sensitivity remains post-hoc welfare or sensitivity accounting: Gamma changes the accounting of unserved requests after simulation and does not alter routing, offer generation, or acceptance. The Beijing-inspired synthetic grid remains a synthetic robustness setting. The algorithm diagnostic, including the simplified ex-post diagnostic over fixed accepted sets, is used to describe method scope and limitations, not to establish a complete dynamic routing benchmark.

7.5. Phase 4 Numerical Handoff

Phase 4 will decide which final tables, figures, and numerical sentences remain in the main manuscript. The handoff is:

- primary behavioral claims must trace to formal Phase 6 main behavioral outputs;
- diagnostic coverage-confounding control claims must stay diagnostic;
- fixed-accepted-set decomposition claims must name the fixed accepted-set scope;
- robustness and passenger-type claims must stay conditional and limited;
- Gamma claims must stay post-hoc;
- algorithm diagnostic claims must stay scoped to simplified fixed-set diagnostics.

Any final value inserted in Phase 4 must use the claim ledger provenance fields: source path, script path, generation command, metric formula, numerator, denominator, evidence role, allowed sentence, and prohibited sentence. Until then, this section uses non-numeric wording and records the evidence families that require Phase 4 verification.

8. Managerial and Operational Implications

The evidence in Section 7 is best read as tested simulation evidence for DRT service design. It does not provide deployment rules or a production planning tool. The implications below translate the model into bounded design considerations for operators who are deciding whether and how to test bidirectional meeting-point service.

8.1. Walking Tolerance as a Design Consideration

Walking tolerance is the first constraint on bidirectional meeting-point consolidation. When candidate pickup and dropoff points require too much walking, the binary-logit response layer shifts otherwise feasible offers into choice rejection. Operators should therefore treat walking-radius settings as local calibration parameters rather than fixed rules. A practical pilot should observe pedestrian network quality, stop spacing, weather exposure, safety conditions, and user willingness to walk before expanding the service design.

8.2. Fleet Deployment and Coverage Risk

Fleet deployment affects whether meeting-point consolidation can translate into useful service. Additional vehicles can reduce infeasibility and waiting burdens, but the passenger-response layer can still constrain served share when walking or delay burdens remain high. Operators should therefore evaluate vehicle-km per served trip together with vehicle-km per original request, served share, choice rejection, and feasibility rejection. A lower routing intensity signal is operationally meaningful only when the coverage and rejection consequences are visible.

8.3. Consolidation Trade-Off

Bidirectional pickup and dropoff consolidation is not an unconditional replacement for door-to-door service. It is a service-design option for conditions in which accepted requests are spatially compatible and passenger burdens remain acceptable. Door-to-door, single-sided pickup, and bidirectional service can coexist as operating modes. The managerial question is which mode fits a demand density, fleet supply, walking tolerance, and reliability target, not whether one mode is globally dominant.

8.4. Passenger-Segment Monitoring

The passenger-type results should be used as monitoring evidence for simulated heterogeneity. They suggest that operators should inspect acceptance and burden patterns by passenger segment when piloting meeting-point service. This monitoring can support fallback design, opt-in rules, fare experiments, or differentiated service offers, but it does not establish field equity effects. Passenger-segment monitoring should be paired with empirical calibration before it is used for external reporting.

8.5. Validation Limits and Diagnostic Evidence

Several evidence layers support interpretation without becoming headline evidence. Matched coverage is a diagnostic coverage-confounding control. Fixed accepted sets are a diagnostic decomposition over fixed accepted sets. Gamma outputs are post-hoc welfare or sensitivity accounting and do not alter routing, offer generation, or acceptance. The Beijing-inspired synthetic grid is a tested simulation setting, not field validation. The MILP comparison is a simplified ex-post diagnostic over fixed accepted sets, not a complete dynamic routing standard. These validation limit statements should remain with any operational interpretation of the findings.

8.6. Calibration-Dependent VOT Interpretation

The value-of-time interpretation of the utility coefficients is calibration-dependent. The coefficients define the simulated passenger-type sensitivities used in this study; they should not be interpreted as validated values for a particular population. Operators who want to use the framework for local planning should estimate or stress-test walking, waiting, in-vehicle time, and fare sensitivities from local stated-preference, revealed-preference, or pilot acceptance data before drawing operational conclusions.

9. Conclusion

This paper studied passenger-response-aware bidirectional meeting-point DRT service design for many-to-many operations. The framework links candidate pickup and dropoff meeting-point generation, single-offer binary-logit passenger response, and rolling-horizon dispatch. It is designed to evaluate operational trade-offs: whether meeting-point consolidation can reduce routing intensity for served trips under tested synthetic service-design conditions, and what happens to coverage, choice rejection, feasibility rejection, and simulated passenger-type heterogeneity when that consolidation is introduced.

The contribution is conditional. Under the tested simulation settings, bidirectional meeting-point consolidation can produce lower routing intensity among served trips, but this signal must be interpreted with the accompanying coverage and passenger-type trade-offs. The full passenger-response-aware design is therefore not an unconditional substitute for door-to-door service. It is an operational design option whose usefulness depends on demand density, fleet deployment, walking tolerance, and the quality of offers presented to passengers.

The paper also clarifies evidence boundaries for future manuscript and package work. Primary behavioral claims should be based on formal Phase 6 evidence and should report the relevant denominators. Matched-coverage, fixed-accepted-set, robustness, passenger-type, Gamma, Beijing-inspired, and algorithm materials are diagnostic, robustness, monitoring, or limitation evidence unless Phase 4 verifies stronger source-specific wording. Gamma remains post-hoc welfare or sensitivity accounting. The Beijing-related evidence remains synthetic. The MILP remains a simplified ex-post diagnostic over fixed accepted sets.

Several limitations remain. The passenger-response parameters are simulation inputs, not empirical calibration from real users. The scenarios are synthetic or Beijing-inspired rather than public-data validations. Passenger types represent service-sensitivity ranges, not real demographic groups. The routing diagnostic is simplified and fixed-set, not a complete dynamic exact model. These limitations are important because they define what the current evidence can and cannot support.

Future work should address these boundaries directly. Real public-data ingestion and field calibration would support empirical passenger-response estimates. Endogenous Gamma behavior would require a model in which welfare accounting affects offer or routing decisions by design. Stronger exact methods would be needed to compare the full online behavioral process rather than a fixed accepted set. More detailed passenger data would also allow passenger-type monitoring to be replaced by empirically grounded equity and burden analysis.

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Appendix A. Notation and Units Reference

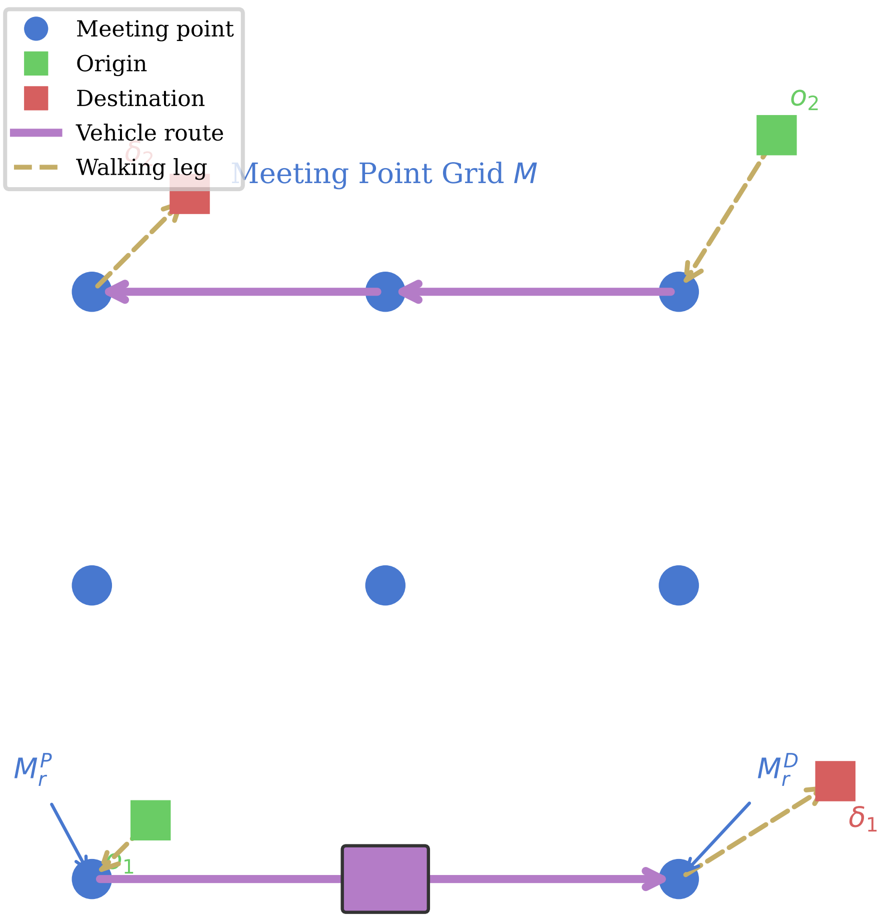


Figure 1: System overview: many-to-many DRT with bidirectional meeting points. Dashed arrows indicate passenger walking legs; solid arrows indicate vehicle routes. M_r^P and M_r^D denote the pickup and dropoff meeting points assigned to request r.

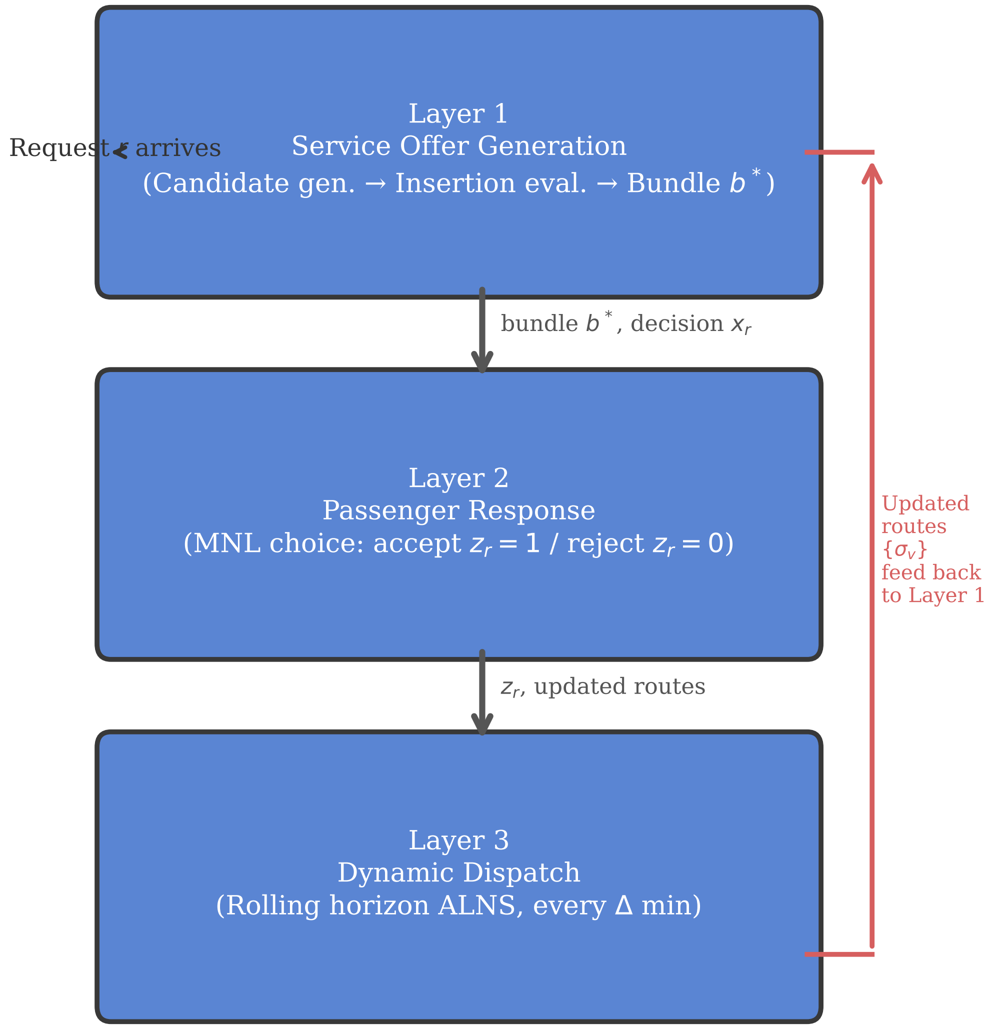
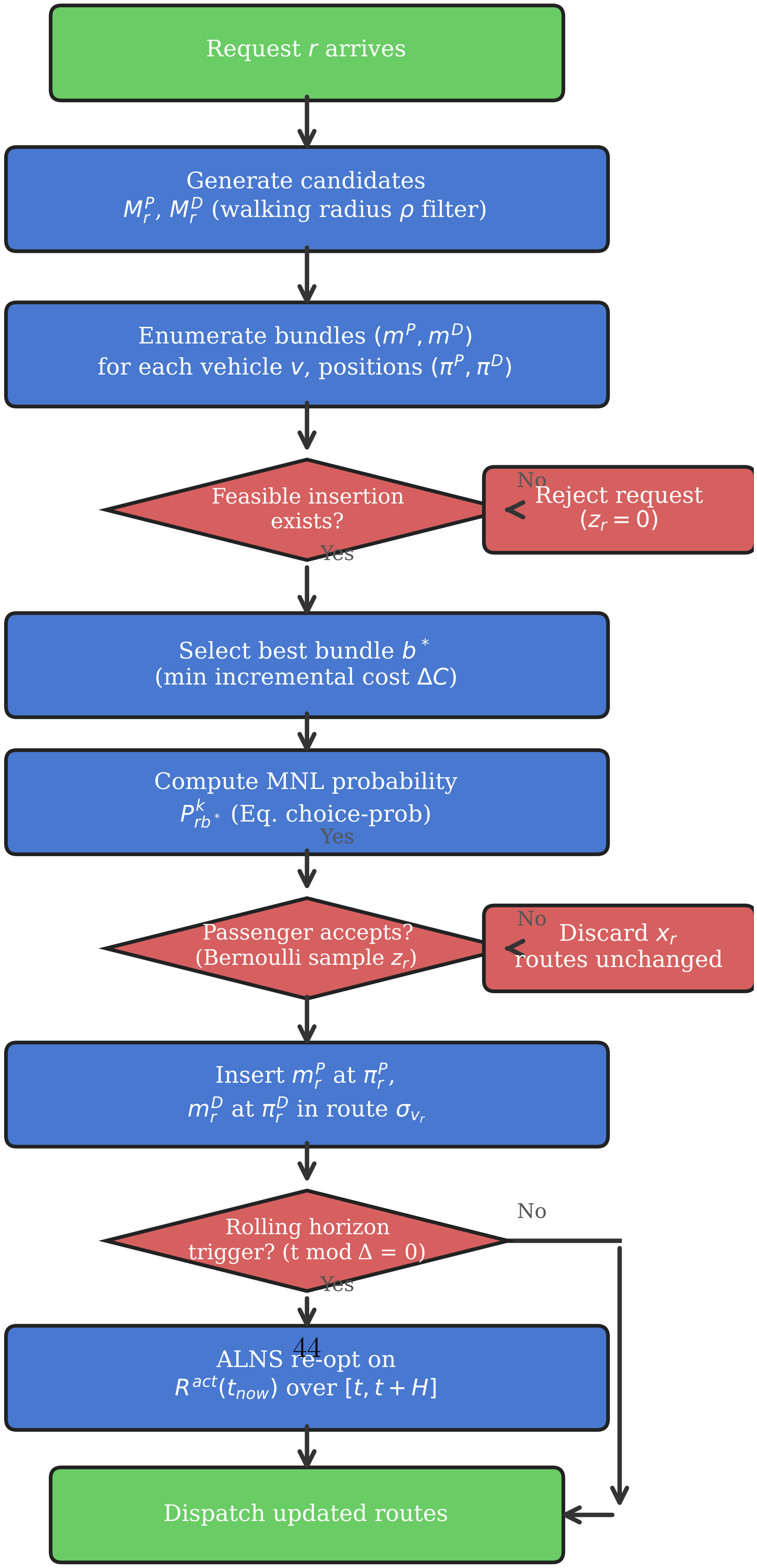


Figure 2: Three-layer coupled model architecture for many-to-many DRT with bidirectional meeting points.


Request r arrives
Generate candidates
M_r^P, M_r^D (walking radius ρ filter)
Enumerate bundles (m^P, m^D)
for each vehicle v, positions (π^P, π^D)
Feasible insertion exists?
No
Reject request
$(z_r = 0)$
Yes
Select best bundle b^*
(min incremental cost ΔC)
Compute MNL probability
$P^k_{rb^*}$ (Eq. choice-prob)
Yes
Passenger accepts?
(Bernoulli sample z_r)
No
Discard x_r
routes unchanged
Insert m_r^P at π_r^P,
m_r^D at π_r^D in route σ_{v_r}
Rolling horizon trigger? (t mod Δ = 0)
No
Yes
ALNS re-opt on
$R^{act}(t_{now})$ over $[t, t+H]$
Dispatch updated routes

Algorithm 1 Rolling Horizon ALNS for Bidirectional DARPmp

Require: Request stream $\mathcal{R}$, fleet $\mathcal{V}$, meeting points $\mathcal{M}$, horizon H, re-opt interval Δ, rejection penalty Γ

Ensure: Accepted set $\mathcal{R}^{\text{acc}}$, routes $\{\sigma_v\}_{v \in \mathcal{V}}$

Initialize routes $\sigma_v \leftarrow \emptyset$ for all $v \in \mathcal{V}$
$t^{\text{now}} \leftarrow 0$
while request stream not exhausted **do**
    // — **Online dispatch: process next arriving request** —
    Receive request r at time t_r^{req}
    $M_r^P \leftarrow \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$, $M_r^D \leftarrow \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$ ▷ Candidate meeting points
    $b^* \leftarrow \textsc{BestInsertion}(r, M_r^P, M_r^D, \{\sigma_v\})$ ▷ Layer 1: select best feasible bundle
    if $b^* = \emptyset$ **then**
        $z_r \leftarrow 0$; record rejection ▷ No feasible insertion found
    else
        $p_{\text{acc}} \leftarrow \dfrac{\exp(U_{rb^*}^{k_r})}{1 + \exp(U_{rb^*}^{k_r})}$ ▷ Binary logit acceptance (Eq. 24)
        $z_r \sim \text{Bernoulli}(p_{\text{acc}})$ ▷ Layer 2: passenger response
        if $z_r = 1$ **then**
            Insert b^* into $\sigma_{v_{b^*}}$ at positions (π^P, π^D)
            $\mathcal{R}^{\text{acc}} \leftarrow \mathcal{R}^{\text{acc}} \cup \{r\}$
        end if
    end if
    // — **Periodic re-optimization** —
    if $t^{\text{now}} \bmod \Delta = 0$ **then**
        $\mathcal{R}^{\text{act}} \leftarrow \{r' \in \mathcal{R}^{\text{acc}} \mid r' \text{ not yet completed}\}$
        $\{\sigma_v\} \leftarrow \text{ALNS}(\mathcal{R}^{\text{act}}, \{\sigma_v\}, H)$ ▷ Layer 3: rolling horizon re-opt
    end if
    $t^{\text{now}} \leftarrow t^{\text{now}} + \delta t$
end while
return $\mathcal{R}^{\text{acc}}$, $\{\sigma_v\}$

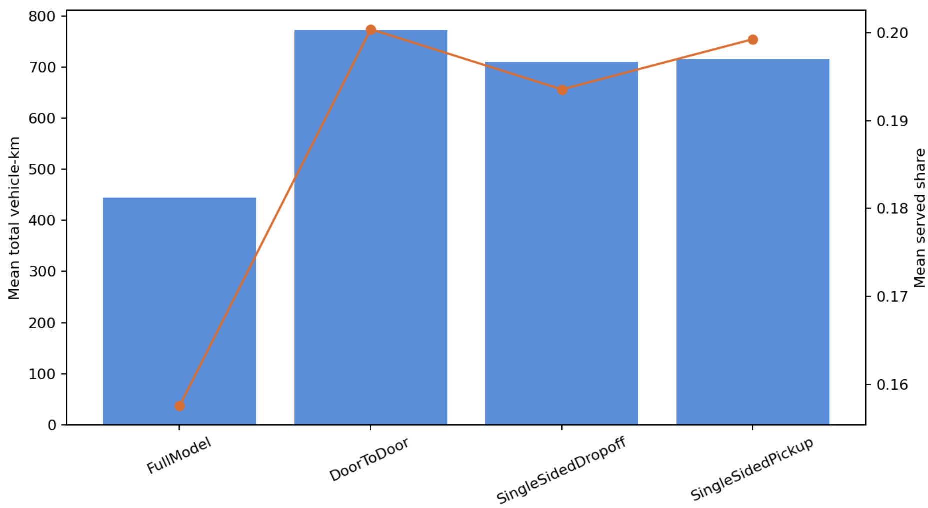


Figure 4: Formal Phase 6 main behavioral summary of routing intensity and coverage. The figure is generated by the formal closeout workflow from the formal main behavioral table and copied from the plot artifact recorded in the formal plot metadata.

Table A.7: Notation and units reference

Symbol	Description
r	Passenger request index
v	Vehicle index
m_r^P	Pickup meeting point for request r
m_r^D	Dropoff meeting point for request r
M_r^P	Candidate pickup meeting point set
M_r^D	Candidate dropoff meeting point set
b_r	Service offer bundle $(m_r^P, m_r^D, \tau_r, p_r)$
τ_r	Scheduled pickup time (internal: seconds; utility: converted to minutes)
p_r	Fare offered to passenger r
ρ^P, ρ^D	Pickup/dropoff walking radius (unified as ρ when $\rho^P = \rho^D$)
H	Rolling horizon window length
Δ	Re-optimization interval
Γ	Rejection penalty in social welfare metric (post-hoc only; not in routing objective)
$U_r(b)$	Deterministic utility of bundle b for request r
$P_{\text{accept}}(b)$	Binary logit acceptance probability
z_r	Realized acceptance indicator ($z_r \sim \text{Bernoulli}(P_{\text{accept}})$), not an optimized routing deci
β_1^k	Walk disutility coefficient for type k
β_2^k	Wait disutility coefficient for type k
β_3^k	IVT disutility coefficient for type k
β_4^k	Price disutility coefficient for type k
k^{top}	Number of candidate meeting points retained per request side
$\bar{v}$	Average vehicle travel speed (30 km/h = 8.33 m/s)
$\mathcal{R}^{\text{acc}}$	Set of accepted requests
$\mathcal{R}^{\text{act}}(t)$	Active request set at time t
σ_v	Route sequence for vehicle v